Process Characterization Plan

A-Mab Drug Substance — Ultrafiltration / Diafiltration (Step 10) — PCP-010

Process Development / Manufacturing Science & Technology (MSAT)

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Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
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Approver	Quality Assurance	—	—	—

Abbreviations

ALCOA+, attributable, legible, contemporaneous, original, accurate, and complete, consistent, enduring, available; AMV, analytical method validation; CEX, cation exchange chromatography; CQA, critical quality attribute; DS, drug substance; DV, diavolume; HMW, high molecular weight species; icIEF, imaged capillary isoelectric focusing; KPP, key process parameter; MSAT, Manufacturing Science and Technology; NOR, normal operating range; PAR, proven acceptable range; PCP, Process

Characterization Plan; PCR, Process Characterization Report; SEC, size exclusion chromatography; SOP, standard operating procedure.

1 Purpose and scope

This plan defines the process characterization study for the ultrafiltration and diafiltration step of the A-Mab drug substance process. The step concentrates the virus-filtered pool, exchanges it into the formulation buffer, and delivers the drug substance at its target concentration of 75 g/L. It is the last unit operation of the drug substance process, and the material it delivers is the material that is filled and released.

The step forms no critical quality attribute, and no clearance claim is made for it. Its parameters are key process parameters (KPP), and the purpose of characterizing them is to bound process performance and batch consistency and not to control a quality attribute. That changes what this study has to establish, because everything it has to establish is a property of the operation itself. Buffer exchange must be complete, the target concentration must be met, the product mass must be accounted for, and the product must leave the step in the state in which it entered.

The plan covers the scale-down system, the 3 parameters listed in Table 6.1, the process performance responses associated with them, and the confirmatory monitoring of the two drug substance attributes (aggregate content and charge variants) that a concentration operation could plausibly affect. The study will be executed at Novacyte Biologics on a qualified scale-down model, under the operating procedure for the step (SOP-2013) and the scale-down qualification procedure (SOP-1001), in support of the technology transfer described in PTP-001.

Formulation characterization is out of scope. The identity and composition of the diafiltration buffer are set by the drug product development programme, and they are controlled at this step by SOP-2013. Excipient levels, pH, viscosity and the stability of the formulated drug substance are characterized and reported in the drug product package, and this plan takes the formulation as given. Commercial-scale process performance qualification is also out of scope, and the 5 qualification batches are planned separately under PTP-001.

2 Objectives

The study has five objectives.

1. Qualify the scale-down ultrafiltration and diafiltration system against commercial-scale performance, so that ranges established on it can be applied to the commercial process.
2. Quantify the effect of each parameter in Table 6.1 on the process performance responses (buffer exchange, final concentration, step yield and mass balance) across the characterization ranges.
3. Establish a proven acceptable range for each parameter against the acceptance criteria of Section 7.
4. Confirm that aggregate content and charge variant distribution are unchanged across the step, within the precision of the methods used to measure them (AMV-3011 and AMV-3013).
5. Confirm the key process parameter classification of all 3 parameters, and provide the operating ranges that the control strategy will carry into Stage 2.

3 Related documents

This plan sits inside the characterization package summarized in Table 3.1. Its parents define the transfer scope (PTP-001), the risk basis for the study scope (RA-001) and the campaign-level approach (PCMP-001). The executed study will be reported in PCR-010 and will roll up into PCMR-001.

Table 3.1: Related documents in the A-Mab characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-010	Process Characterization Report (Ultrafiltration / Diafiltration (Step 10))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The controlled procedures and validated analytical methods that govern the study are listed in Table 3.2. All numbers in that table are placeholders for a synthetic document set.

Table 3.2: Controlled procedures and validated methods for the step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3019	Protein Concentration by A280 (UV)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The step is operated as tangential flow filtration in three phases. The virus-filtered pool is first concentrated against a retentive membrane, then diafiltered at constant volume against the formulation buffer, and finally concentrated to the target before the system hold-up volume is recovered by a buffer flush. Product is retained by the membrane throughout, and buffer species pass into the permeate.

Retention is by size, and for an antibody it is close to quantitative. Monomer, aggregate, charge variants and glycoforms are retained together, so the step concentrates the product without fractionating it. This is the mechanistic reason that the step forms and clears no quality attribute, and it is why the parameters of the step are classified for process performance.

Some low molecular weight species pass into the permeate. No impurity clearance is claimed for this step, because the clearance that matters is delivered upstream and is credited there. Host cell protein, residual DNA and leached Protein A are cleared by the chromatographic steps and are reported in PCR-005, PCR-007 and PCR-008. No viral clearance claim is made for this step either. The modular clearance claim rests on the low-pH hold, the anion exchange step and the small virus retentive filtration step (PCR-006, PCR-008 and PCR-009), and it is consolidated in PCMR-001 (International Council for Harmonisation 2023).

Ultrafiltration and diafiltration is a platform operation at Novacyte Biologics. The same membrane chemistry, the same crossflow and pressure window, and the same diafiltration strategy have been used for three earlier humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and throughout A-Mab clinical manufacture, and the membrane retained all of those products quantitatively, so no product-specific difference in retention is expected for A-Mab. This prior knowledge is why the step is characterized univariately and at a smaller scale of effort than the chromatographic steps (CMC Biotech Working Group 2009).

Three mechanisms are known from that platform experience to matter, and the study is designed around them. Incomplete buffer exchange at a low number of diavolumes leaves residual feed buffer in the drug substance, which shifts the conductivity and osmolality of the final pool away from the values of the formulation buffer and is seen in process before the batch is filtered into its container. High transmembrane pressure raises the protein concentration at the membrane surface, which is the condition under which aggregation is most likely during a concentration operation (the size-variant assay AMV-3011 is the readout). A high final concentration raises viscosity, lengthens the final concentration phase, and increases both the shear history of the product and the fraction of the batch retained in the system. Each of these acts through hydraulics and mass balance, and none of them acts through the chemistry that forms a quality attribute.

4.2 Quality attributes in scope

No drug substance attribute is formed or cleared at this step, so the step is assigned none in the quality attribute register. Two attributes are nevertheless monitored across it, and they are given in Table 4.1. Aggregate content is monitored because concentration and shear are the plausible routes by which a size variant could be generated this late in the process. Charge variants are monitored because the product is held in the formulation buffer at the end of the step, and deamidation is a pH- and time-dependent reaction.

Table 4.1: Drug substance attributes monitored across the step. Neither is set nor cleared here; both are measured to confirm that the step does not change them.

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4

Both attributes are set upstream. Aggregate is formed in the production bioreactor (PCR-003) and reduced principally by cation exchange (PCR-007), and it carries high criticality. Acidic charge variants are also formed upstream and are of very low criticality. The acceptance criteria in Table 4.1 are the drug substance criteria that the final pool has to meet in any case, so the study measures a change across the step against them and does not set a limit of its own.

Glycosylation, host cell protein, residual DNA, leached Protein A and viral clearance are not measured across this step, because a size-based concentration and buffer exchange operation has no mechanism by which it could alter them. Each of them is already characterized in the report of the step that governs it (PCR-003 for the glycans, PCR-005 to PCR-009 for the impurities and the viral clearance).

4.3 Risk-based prioritization of parameters

RA-001 ranked the parameters of this step before the study was designed and assigned all of them to univariate assessment. The ranking is driven by its severity axis. Because no critical quality attribute is at risk here, no parameter of the step scores high enough to require a multivariate design, and the interaction ranking that would otherwise justify one is low as well. The risk these parameters carry is to process performance and to batch-to-batch consistency, which is what makes them key process parameters and not critical process parameters (CMC Biotech Working Group 2009).

The characterization ranges in Table 6.1 are wider than the normal operating ranges, so that the study bounds excursions as well as routine operation. The diavolume and pressure ranges extend above the normal operating range and share its lower edge, and the final concentration range extends beyond the normal operating range in both directions. The low edge of the diavolume range is the edge at which buffer exchange becomes measurably incomplete, and it is included for that reason. The pressure range is extended upward because that is the direction in which concentration polarization and aggregation risk increase, and the final concentration range is challenged on both sides, because its low edge carries a fill volume consequence and its high edge carries the viscosity and aggregation consequence.

5 Materials and methods

5.1 Scale-down model and its qualification

The study will be run on a scale-down tangential flow filtration system that reproduces the commercial step on the variables that govern its performance. The model uses the same membrane chemistry and channel geometry as the commercial cassettes, and it is operated at the same membrane loading, the same crossflow per unit channel height, the same transmembrane pressure and the same number of diavolumes. One scale-dependent quantity cannot be matched. The ratio of system hold-up volume to batch volume is larger at small scale, and its consequence for the study is stated in Section 12.

Qualification will follow SOP-1001. The set-point condition will be run in 3 replicates, and the flux profile, the step yield, the final concentration and the two monitored attributes will be compared with the corresponding commercial-scale records. The model will be accepted as qualified when the scale-down values agree with the commercial-scale values within the reproducibility band established by those replicates, and when the flux profile has the same shape. The qualification data will be reported in PCR-010 alongside the study results.

5.2 Operation

The feed to the step is the filtrate of the small virus retentive filtration step (Step 9, PCR-009). Every run will follow SOP-2013, with only the parameter under study moved away from its set-point. The pool is concentrated to an intermediate concentration, diafiltered at constant volume against the formulation buffer for the number of diavolumes set for the run (Table 6.2), and then concentrated to the final target, after which the system is flushed with formulation buffer to recover the hold-up volume. The recovered material is pooled with the retentate and filtered into the drug substance container.

Membrane lot, buffer lots and system hold-up volume are controlled to platform specification under SOP-2013. They are identity and quality controlled inputs to the study, and they are not characterized by it.

5.3 Analytical methods

Table 3.2 lists the validated methods. Protein concentration is measured by absorbance at 280 nm (AMV-3019), and it is the basis of both the final concentration result and the mass balance. Size variants are measured by size exclusion chromatography (AMV-3011), and charge variants by imaged capillary isoelectric focusing (AMV-3013). Conductivity and osmolality are in-process measurements taken under SOP-2013, and they are the direct readout of buffer exchange.

Method performance is defined in the validation reports and is not restated here. The intermediate precision of AMV-3019 is 1.2 % relative standard deviation, which sets the resolution at which a difference in final concentration or in recovered mass can be claimed. The corresponding figures for the size-variant and charge-variant methods are given in AMV-3011 and AMV-3013.

5.4 Sampling plan

Samples will be taken at five points in every run. These are the feed pool, the end of the concentration phase, each whole diavolume of the diafiltration, the final concentrate before the recovery flush, and the pooled drug substance after it.

Conductivity and osmolality will be measured on every diavolume sample, so that the approach of the retentate to the formulation buffer is followed across the whole diafiltration and not read off its endpoint alone. Protein concentration (AMV-3019) will be measured on every sample and on the pooled permeate and flush volumes, which is what closes the mass balance, while size variants (AMV-3011) and charge variants (AMV-3013) are measured on the feed pool and on the pooled drug substance of every run. The intermediate samples of the worst-case run are analysed as well.

5.5 Statistical methods

The planned analysis is descriptive, because no response model will be fitted at this step. Each parameter is varied one at a time with the other two at their set-points, so the design supports a statement about each parameter on its own but not about interactions between them, and the combined run of Section 6 is the only condition at which more than one parameter moves. The set-point replicates give the reproducibility of every response as a mean and a coefficient of variation, and that band is the reference against which each edge condition is judged.

A parameter will be reported as having an effect on a response when the value at the edge condition falls outside the set-point band and the difference is larger than the precision of the method used to measure it. Change in a monitored attribute across the step will be judged the same way. With 3 replicates the reproducibility estimate is itself imprecise, and the study is not powered to detect a small effect. It is powered to detect the effects that matter here, which are incomplete buffer exchange and a loss of product mass, and both of those are large when they occur.

No commercial-scale capability estimate will be produced for this step. Capability is estimated for the attributes that a step governs, and this step governs none. The Monte-Carlo estimates over 2,000 simulated batches are reported in the documents of the steps that set each attribute, and they are consolidated in PCMR-001.

6 Study design

6.1 Factors, ranges and study type

Table 6.1 gives the parameters, their set-points, the ranges to be studied, the normal operating ranges and the study type assigned to each. All of them are assigned to univariate assessment, and no parameter of this step is carried into a multivariate design.

Table 6.1: Parameters to be studied, with set-points, characterization ranges and normal operating ranges.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Number of diavolumes	DV	8	7–10	7–9	univariate
Transmembrane pressure	psi	12	10–15	10–14	univariate
Final DS concentration	g/L	75	65–85	70–80	univariate

6.2 Univariate assessment

No screening design and no response-surface design is planned for this step. There is no quality attribute response to model, and the 3 parameters act through hydraulics and mass balance and not through a reaction whose rate they could jointly control, which is the condition under which a multivariate design earns its cost. The designs used at the bioreactor and at the chromatographic and filtration steps (PCP-003, PCP-005 to PCP-009) are therefore not applied here, and no design space will be claimed for this step. The outcome of the study is a set of proven acceptable ranges (PAR) and a confirmed classification, which is what a key process parameter needs.

The planned run list is given in Table 6.2 and contains 10 runs. Each parameter is run at both edges of its characterization range with the other two at their set-points, the set-point condition is run in 3 replicates, and one further run combines the worst case of all 3 parameters.

Table 6.2: Planned univariate run list. Levels are the set-points and the characterization-range edges of Table 6.1.

Run	Number of diavolumes (DV)	Transmembrane pressure (psi)	Final DS concentration (g/L)	Condition
1	7	12	75	Number of diavolumes low edge
2	10	12	75	Number of diavolumes high edge
3	8	10	75	Transmembrane pressure low edge
4	8	15	75	Transmembrane pressure high edge
5	8	12	65	Final DS concentration low edge
6	8	12	85	Final DS concentration high edge
7	8	12	75	Set-point replicate
8	8	12	75	Set-point replicate
9	8	12	75	Set-point replicate
10	7	15	85	Combined worst case

The combined run is the worst case for this step. It sets pressure and final concentration at their high edges and the number of diavolumes at its low edge, so aggregation risk and hold-up losses are at their highest while buffer exchange is at its least complete. Meeting the acceptance criteria at that condition, as well as at every single-parameter edge, is what allows the univariate ranges to be read together with some confidence. It is a check on independence and it is not an interaction estimate.

7 Acceptance and decision criteria

The study passes when the scale-down model is qualified, when the four process performance criteria below are met across the characterization ranges, and when the two monitored attributes are unchanged across the step and within the drug substance criteria of Table 4.1. Each criterion is stated here, before any data exist.

Buffer exchange is the first criterion. Retentate conductivity and osmolality at the end of diafiltration must lie inside the control range of the formulation buffer (SOP-2013) at every diavolume level studied. Under ideal constant-volume diafiltration with complete mixing, the residual fraction of a freely permeating feed-buffer species falls exponentially with the number of diavolumes, so that at the low edge of the characterization range the ideal residual is 0.09 % of its starting value and at the set-point it is 0.03 %. Those values are the basis on which the low edge of the range was set. The study will measure the actual approach to the formulation buffer and will not assume it.

Final concentration is the second criterion. The concentration of the pooled drug substance must lie inside the normal operating range of the final concentration parameter, which is 70–80 g/L, when the step is run at its set-point. At the edge conditions the requirement is that the target set for the run is achieved within the precision of AMV-3019.

Step yield and mass balance are the third criterion. Yield at each condition will be compared with the platform expectation of 96.0%, whose batch-to-batch coefficient of variation is 1.5%. A condition at which yield falls short of that expectation by more than the set-point reproducibility will be investigated before its range is accepted. The mass balance of every run must close, with the product mass in the retentate, the permeate and the flush accounting for the mass in the feed within the precision of AMV-3019.

Product quality is the fourth criterion. Aggregate content and charge variant distribution of the pooled drug substance must meet the criteria of Table 4.1 at every condition, and the change across the step must not exceed the precision of AMV-3011 and AMV-3013 respectively, measured between the feed pool and the final pool of the same run. A larger change at any condition will be treated as an effect of the step and investigated, whether or not the pool still meets the drug substance criterion.

A condition that fails one of these criteria does not fail the study. It bounds the parameter, and the proven acceptable range is then reported as the part of the characterization range over which the criteria are met (Section 8). A failure at the set-point condition is different, and so is a failure of the scale-down model qualification. Either one stops the study, and both are handled as deviations under the campaign procedures of PCMP-001.

8 Proven acceptable ranges (planned analysis)

The acceptance basis is stated before the analysis, and it differs by response. For the two monitored attributes the basis is the drug substance acceptance criterion in Table 4.1. For buffer exchange, final concentration, yield and mass balance the basis is the process performance criteria of Section 7, because those responses are not quality attributes and carry no specification of their own. No step-level clearance requirement is back-calculated for this step, since it makes no clearance claim.

One analysis will be run per parameter. Each parameter will be assessed across its characterization range at the levels given in Table 6.2, with the other two held at their set-points, and its proven acceptable range will be reported as the widest continuous part of that range over which every criterion is met. Where the whole characterization range is acceptable, the report will say so, and the proven acceptable range of that parameter is then its full studied range.

The NOR-propagated analysis used at the multivariate steps will not be run here. That analysis varies the other factors within their normal operating ranges by Monte-Carlo simulation of a fitted response-surface model, and no such model will exist for this step. The combined worst-case run of Section 6 is what this plan offers in its place.

Each proven acceptable range will be presented as a table of the measured response at every level studied against its acceptance limit, with the acceptable region marked. No fitted curve will be shown, because none will be fitted. The report must state the two bounds on these ranges. They hold with the other two parameters at their set-points, and they hold on the qualified scale-down model, whose hold-up volume fraction differs from that of the commercial system.

9 Data management and integrity

Raw data will be recorded contemporaneously in the validated electronic laboratory notebook, and analytical results in the chromatography data system and the laboratory information management system. Records will meet the ALCOA+ expectations (attributable, legible, contemporaneous, original and accurate) set out in the site data integrity procedures. Instrument audit trails will be enabled and reviewed as part of the data review.

Analysis scripts and their inputs will be version controlled and archived with the report, so that every table in PCR-010 can be regenerated from the raw data. A second person will review every calculation before the report is issued.

10 Roles and responsibilities

Process Development and MSAT own this plan, execute the runs and author PCR-010. Analytical Development performs the assays and owns the method performance data that the acceptance criteria of Section 7 depend on. Manufacturing Science reviews the scale-down model and the applicability of the resulting ranges to the receiving site. Statistics reviews the reproducibility estimate and the analysis. Quality Assurance approves the plan and the report and dispositions any deviation. The approval record is Table 1.

11 Deliverables and schedule

The deliverable of this plan is PCR-010. It will report the qualification of the scale-down model, the results of the 10 planned runs, the proven acceptable range of each parameter, the confirmed classification of all 3 parameters (KPP), and the contribution this step makes to the control strategy.

The work runs in four phases. The scale-down model is qualified first, the runs are then executed, the analysis follows, and PCR-010 is issued last. PCR-010 must be approved before the ranges of this step are carried into the commercial batch record and before the qualification batches described in PTP-001 are executed. No dates are given here, and the campaign schedule is held in PCMP-001.

12 Risks and assumptions

The principal risk to the study is the representativeness of the scale-down model, and it comes from one quantity. The fraction of the batch held in the system hold-up volume is larger at small scale than at commercial scale, so the scale-down step yield is expected to be lower than the commercial yield and to be more sensitive to the recovery flush than the commercial step is. Yield will therefore be assessed as a difference from the scale-down set-point condition and not as an absolute value transferred to commercial scale. The qualification runs of Section 5.1 are what establish that this difference is stable.

Two further risks are recorded. The feed for the study is the pool of the small virus retentive filtration step, so the study depends on the upstream campaign for representative feed material, and any substituted feed would have to be qualified before use. Membrane lot variability is not studied here, and it is controlled by the incoming specification held under SOP-2013.

The plan makes three assumptions. Platform behaviour of the membrane is assumed to transfer to A-Mab, which the prior products and A-Mab clinical manufacture support but which no product-specific study at this step has yet confirmed. Diafiltration is assumed to be well mixed and at constant volume, which is the basis of the residual calculation in Section 7 and will be checked against the measured conductivity profile. The formulation composition is assumed fixed, and it is not a variable of this study.

13 Approvals

This plan becomes effective when it is approved by the functions listed in Table 1. Any change to the study design after approval will be recorded as an amendment to this plan and reported in PCR-010 (U.S. Food and Drug Administration 2011).

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

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